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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Catalyst
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 - Decent Work and Economic Growth SDG 9 - Industry, Innovation and Infrastructure
Project / Solution Name	Orbyte SlipMatch - Instant Chat-to-Bill Generator & Automated DuitNow Receipt Validator
Name (Leader)	Ang Yi Xian
Email & Phone (Leader)	Email: xian12053@gmail.com Phone: +6011-1066 1817
University Affiliation(s)	Universiti Malaysia Perlis
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small enterprises (MSMEs) form 96% of Malaysian businesses, almost 1.3 million firms, according to SME Corp. Malaysia, serving as vital economic engines. However, micro-merchants, such as home-based bakers, social media sellers, and artisan retailers operate with lean teams relying heavily on manual, disconnected workflows across WhatsApp, spreadsheets, and personal banking apps. While national digital payment initiatives like DuitNow QR have driven e-payments to 538 transactions per person in 2025, as reported by Bank Negara Malaysia, payment reconciliation remains an exhausting operational bottleneck.

Owners spend hours every evening manually parsing unstructured chat messages, calculating order totals, typing invoices, and cross-referencing payment screenshots against bank account records. This manual process causes severe business friction: frequent arithmetic mistakes, delayed fulfillment, and high vulnerability to payment fraud. Dishonest buyers exploit manual fatigue by submitting doctored screenshots with altered amounts, reusing authentic transaction reference numbers from previous orders, or short-paying balances. Police recorded 14,881 e-commerce fraud cases with RM123.7 million in losses from January to November 2025.

Because micro-enterprises operate on narrow margins and cannot afford enterprise ERP software or commercial payment gateway fees, unrecovered shortfalls directly erode net income. The resulting administrative burden drains productive working hours, induces owner burnout, and stifles business expansion. Solving this recurring burden is critical to supporting sustainable MSME livelihoods, preserving working capital, and accelerating inclusive digital economic growth in Malaysia.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Orbyte SlipMatch is an automation product designed for the Orbyte ecosystem that eliminates operational friction in chat commerce by automating order billing and payment verification while keeping merchants in total control. The system executes a five-stage automated journey: Trigger, Connected Data, Decision Logic, Human Approval, and Visible Outcome.

The journey begins when an unstructured customer WhatsApp order is ingested. Our parser extracts item requirements, matches customer intent against the merchant's dynamic product catalog, generates a sequential order identifier (e.g., ORD-1001), and renders an instant branded payment bill with embedded DuitNow payment credentials. Once the customer transfers funds and uploads a transaction screenshot, SlipMatch initiates automated verification using a high-accuracy Optical Character Recognition (OCR) neural model hosted on cloud infrastructure.

The OCR engine extracts critical transaction metadata, including transfer amount, date, timestamp, recipient name, and alphanumeric bank reference numbers with confidence scoring. SlipMatch's core decision logic cross-references this extracted data against open orders in the database and the day's bank statement. The engine performs combinatorial sum reconciliation (Combo Math) to handle single payments settling multiple invoices, calculates exact-cent shortfalls, and conducts cryptographic hash and reference uniqueness checks to detect duplicate slips.

Crucially, SlipMatch adheres to human-in-the-loop governance: AI never executes irreversible financial confirmations autonomously. Instead, results populate a traffic-light Kanban interface:

- Green indicates verified exact matches
- Yellow flags partial payments or visual ambiguities with exact shortfall calculations
- Red alerts the owner to reused reference numbers or missing bank credits.

Slips below the confidence threshold go to a "Requires Human Eye" queue. The merchant approves or rejects transactions with a single click, triggering automated state transitions and immutable audit logging.

Leveraging skills gained from SDC Phase 1 workshops, our architecture integrates FastAPI backend services, SQLite relational schemas, Replit web deployment, and RunPod Serverless GPU containerization for scalable, low-latency OCR inference

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	**Technology**	**Role**
Frontend + backend	Replit · FastAPI (Python)	Order paste box, upload page, Kanban board, Owner/Staff views
Order Capture AI	GenAI text API / model	Extracts name, items, flavours, date, and total from WhatsApp messages
Vision AI / OCR	RunPod Serverless GPU	Slip reading (amount, date, bank, ref no.) + tamper checks + confidence score
Matching logic	Team-written Python rules	Three-way match (slip, order, bank), Combo-Math, short-pay detection, used-credit check
Database	SQLite	Product catalogue, orders, slip records, audit log
Bank records	Synthetic bank statement (CSV upload)	Proof of what money really arrived
Messaging	WhatsApp click-to-chat link	Pre-filled short-pay message, sent by Owner manually
Access control	Simulated login	Staff / Owner role-based access control (RBAC) demo
Code Respiratory	GitHub	Public repo for judges

System Block Diagram

User Input / WhatsApp Paste → Replit (FastAPI App & UI) → RunPod Serverless Endpoint (OCR Vision AI) → Three-Way Matching Logic & Database → Traffic-Light Kanban / Owner Approval → Audit Log & Outcome

Privacy and PDPA (real deployment)

The demo uses only team-made synthetic data. In real use, slips contain names and account details, so SlipMatch would follow the seven PDPA principles, including retention, keeping data only as long as needed. Since 1 June 2025, amended PDPA rules require data protection officer appointment and mandatory data breach notification. RBAC limits who sees slip data, and the audit log records every access and action.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied human-centered design thinking to empathize with Malaysian micro-merchants. We studied how home bakers and social commerce sellers handle WhatsApp orders and payment slips, utilizing news reports, police fraud data, and SME surveys. We designed for owners who need simple tools and want the final say over money decisions.

In response, we engineered SlipMatch to integrate invisibly into existing merchant habits. The interface requires zero onboarding: order intake mirrors familiar WhatsApp messaging, bill generation produces standard visual receipts, and payment verification is organized into an intuitive traffic-light Kanban board (Green, Yellow, Red).

Technical viability and deployment readiness are ensured through a modular, decoupled architecture. High-frequency API routing and relational state tracking (FastAPI and SQLite) operate independently from compute-intensive OCR neural inference (RunPod Serverless), ensuring responsive performance even during peak sales. Crucially, the system enforces a strict human-in-the-loop boundary, flagging discrepancies and shortfalls for one-click merchant confirmation rather than triggering unmonitored bank updates. By validating our prototype against diverse synthetic slips and establishing clean RESTful interfaces, SlipMatch is structured for immediate deployment as a plug-and-play micro-app within the Orbyte business hub.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: UN SDG Target 8.2 - "Achieve higher levels of economic productivity of economies through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Primary: UN SDG Target 8.3 - "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services." Secondary: UN SDG Target 9.3 - "Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysian micro-enterprises manually spend an average of 12 to 15 hours weekly (accounting for approximately 18% of total productive working hours) transcribing chat orders, calculating manual bills, and verifying payment slip screenshots. In addition, micro-merchants experience an estimated 3.5% to 5.0% monthly revenue loss caused by undetected payment shortfalls, unverified orders, and doctored receipt fraud.

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• Payment Reconciliation Latency: Average time (seconds) required from payment slip image intake to verified invoice match. • Administrative Productivity Gain: Net reduction in manual reconciliation hours per business owner per week. • Fraud & Shortfall Detection Accuracy: Percentage rate of successfully flagged duplicate reference numbers and partial-payment shortfalls.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• 85% reduction in payment reconciliation latency • At least 50% less owner time per slip compared to manual checking • 100% automated detection of reused transaction reference numbers and partial-payment shortfalls across synthetic test suites

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	Code	MIT License
Pydantic	Code	MIT License
HTTPX	Code	BSD License
AnyIO	Code	MIT License

6.3 Employment-Related IP Conflicts

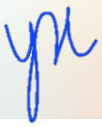
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Ang Yi Xian]

[25 Oct 2026]